u1982583 CH923 Final Module Assessment

Question 1

a) Test for whether the mean concentrations for men and women differ significantly:

Firstly, Do a F test for test of equal variances:

H_0: The Variances between Male and Female albumin concentrations are equal.

$$\sigma_{Female} = \sigma_{Male}$$

H_1: The Variances between Male and Female albumin concentrations are not equal.

$$\sigma_{Female} \neq \sigma_{Male}$$

So under our H_0 : F ~ f_d so we reject H_0 at the 5% level if F >$f_{7,7}(0.95)$

We see that the variances are equal (homoscedasticity of our data), see the p value of 0.9144 > 0.05 (F value of 0.91919 within 3.787044 F test threshold (df1=7,df2=7)) so therefore do not reject H_0 at the 5% level that the variances are equal. Using this information, next is to do the two sample t test which compares the means without a siding and without an alternative hypothesis:

H_0: There is no significant difference in the mean concentration of albumin (gLˆ(-1)) in the blood sera of Males and Females.

$$\mu_{Female} = \mu_{Male}$$

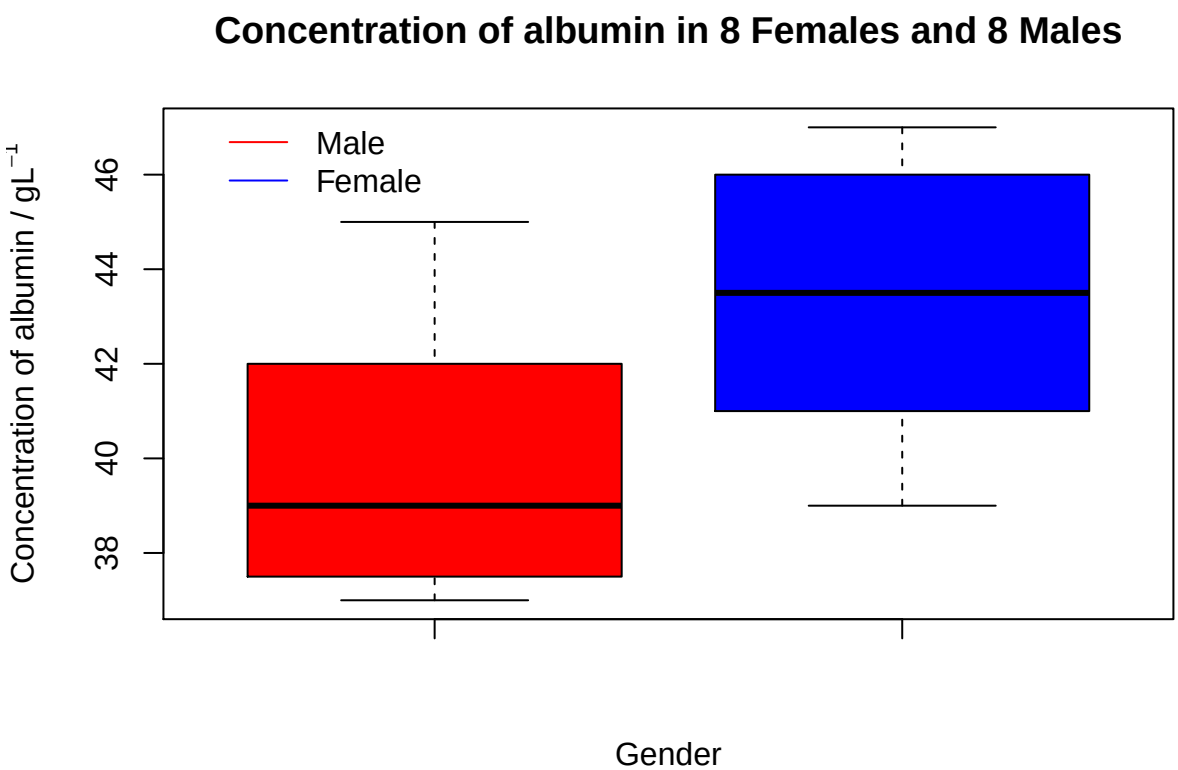
H_1: There is a significant difference in the mean concentration of albumin (gLˆ(-1)) in the blood sera of Males and Females.

$$\mu_{Female} \neq \mu_{Male}$$

Under H_0: T ~ t_d so we reject H_0 at the 5% level if |T|>$t_{14}(0.975)$

with d being m-2=14

```
Male=c(37,39,37,42,39,45,42,38)     #concatenate of albumin concentrations in Males
Female=c(44,40,39,45,47,47,43,42) #concatenate of albumin concentrations in Females
boxplot(Male,Female,xlab="Gender", ylab=expression("Concentration of albumin / gL"^-1),
        main="Concentration of albumin in 8 Females and 8 Males",
        col=c("red","blue","green","orange","brown")
)  #visual representation for inspection.
legend(0.5, 47.5, legend=c("Male", "Female"),
       col=c("red", "blue"), lty=1, cex=1, bty = "n")
```




```r
var.test(Male,Female)  #Variance test for one assumption
```

```
##
##  F test to compare two variances
##
## data:  Male and Female
## F = 0.91919, num df = 7, denom df = 7, p-value = 0.9144
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  0.1840258 4.5912802
## sample estimates:
## ratio of variances
##          0.9191919
```

```r
qf(0.95,7,7)          #Quantile at 95% confidence interval
```

```
## [1] 3.787044
```

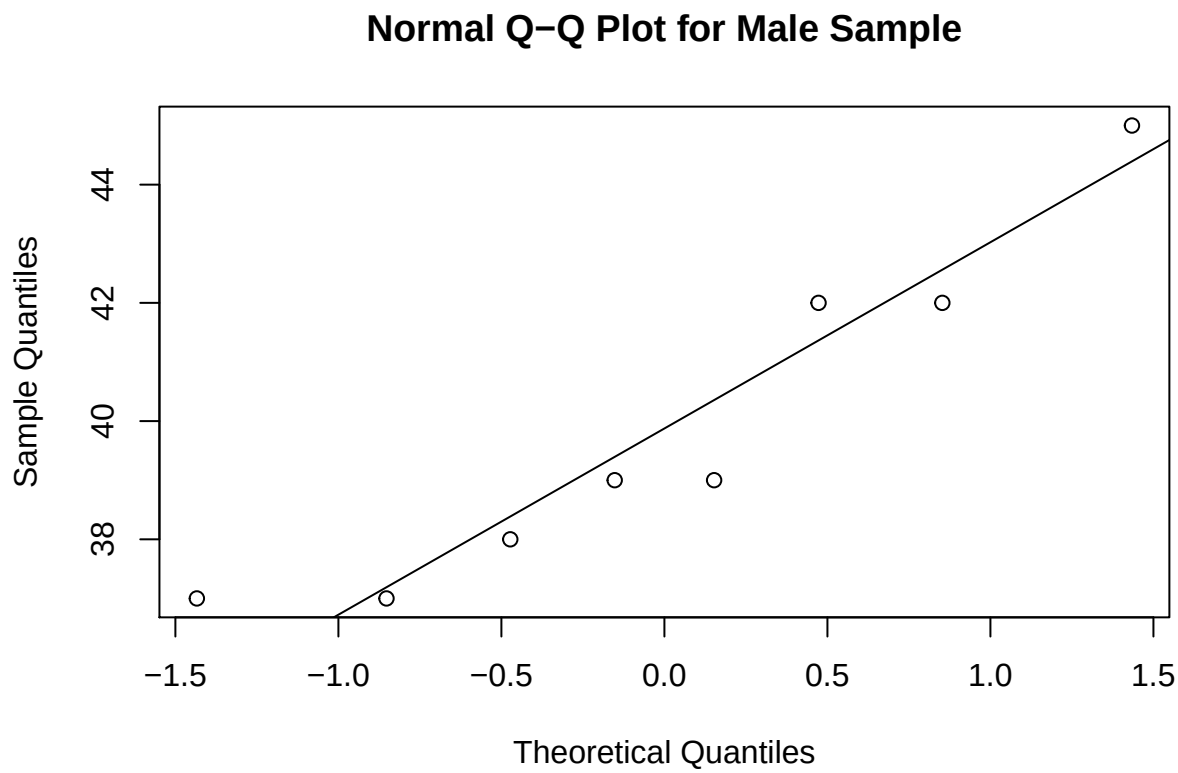
```r
t.test(Male,Female, var.equal = TRUE,paired=FALSE,alternative = "two.sided") #test
```

```
##
##  Two Sample t-test
##
## data:  Male and Female
## t = -2.4035, df = 14, p-value = 0.03066
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -6.6232511 -0.3767489
## sample estimates:
## mean of x mean of y
##    39.875    43.375
```

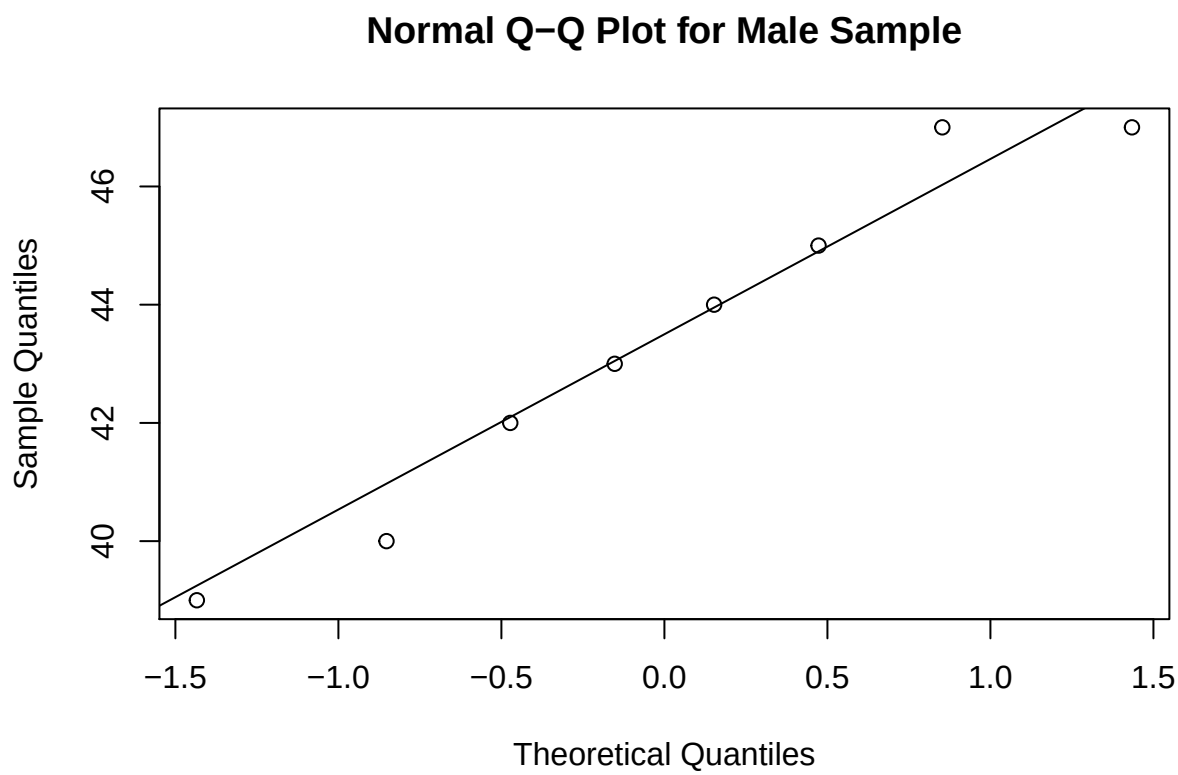
```r
qt(c(0.975,0.025),14) #quantile for the test
```

```
## [1]  2.144787 -2.144787
```

```r
#Test for Assumptions (normality)
qqnorm(Male, main = "Normal Q-Q Plot for Male Sample")
qqline(Male)
```




```r
qqnorm(Female,main = "Normal Q-Q Plot for Male Sample")
qqline(Female)
```




```r
#Quantitative test for the normality assumption
shapiro.test(Male)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  Male
## W = 0.89194, p-value = 0.2439
```

```r
shapiro.test(Female)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  Female
## W = 0.94246, p-value = 0.6355
```

```r
#test for independence,
```

Therefore as T = -2.4035 there was evidence to reject H0 at the 5% level as |T| is bigger than $t_1 5(0.975)$ of 2.144787. There is therefore evidence that Male and Female albumin concentrations are different. The p-value for this test was 0.03066 <0.05. However the number of samples (n) is low and the unclarity of methodology regarding independence of data could render the test invalid.

b) Assumptions necessary for this test.

1. Each population follows a normal distribution (Shapiro test p values of 0.2439 for males and 0.6355 for females >0.05 for males, qqplot for +/- skew shown and satisfied) 2.The observations for each albumin concentration are independent to each other (We are not detailes on the experimental data collection and so assume the 16 healthy adults are not closely related (twins, nuclear family members apart from parents etc) . 3.Samples are from populations with equal variances (Satisfied by variance test).

c) Assessment on validity and reasonability of the assumptions in part b)

Shapiro goodness of fit (W test), test for normality H_0: The Male sample data belongs to a normal distribution.

$$H_0 : Male \in N_o$$

H_1: The Male sample data does not belong to a normal distribution.

$$H_1 : Male \notin N_o$$

H_0: The Female sample data belongs to a normal distribution.

$$H_0 : Female \in N_o$$

H_1: The Female sample data does not belong to a normal distribution.

$$H_1 : Female \notin N_o$$

Reject H_0 if $W_{p-value} < W_\alpha$ Our W_α is set to 0.05

Assumptions are reasonable for these data as shown by the tests (variance and normality) all being within their thresholds, the qq plots show the points lie on the normal line with no skew giving proof sample data is from normal distribution, however the independence of observations is unknown due to the sampling and individuals chosen (family etc), however as are different individuals we can assume independence. Variance is further visually shown to be similar by the width of the boxplpot.

d) Implications of conclusions for doctors interpreting blood test results.

The implications of these conclusions for doctors interpreting blood test results include, flagging gender as a factor leading to higher/lower albumin concentration. Using the same t-test with an alternative of Female > Male gives a T test statistic lower than the 95% threshold and so gives creedance to reject the H_0 at the 5% level. This shows that females have higher albumin concentrations than males, however a low sample size may not be representative for application to the wider population. A factor that is important in the in albumin determinant dignosis including malnutrition, liver disease and inflamatory diseases all shown as having lower albumin levels. Higher albumin levels correlate to by acute infections, burns, stress from surgery or heart attacks. Experimental design would be important in ages of the healthy adults as albumin levels has been shown to decrease with age.[1]

Question 2

a) Estimate ($\hat{\boldsymbol{\theta}}$) for the mean number of individuals infected per household. Calulating the frequncy density total ($= \sum_{i=0}^{5}(x_i * frequency)$) divided by the sample size (60 households) 150/60 = 2.5 $\therefore$ $\lambda = 2.5$

b) Calculation of the probabilities from a Poisson distribution with mean (λ) gives an outcome of 0, 1, 2, 3, 4 and 5. The Poisson distribution has probability mass function e-*lambda lambda*

Mathematical proof:

The Poisson distribution has probability Mass function:

$$p_X(\kappa) = P(X = \kappa) = \frac{e^{-\lambda}\lambda^\kappa}{\kappa!}$$

for κ=0,1,2... Our κ values are 0,1,2,3,4,5 If X=κ then:

P(X =0) $=\frac{e^{-2.5}2.5^0}{0!} = 0.0820...$ P(X =1) $=\frac{e^{-2.5}2.5^1}{1!} = 0.2052...$ P(X =2) $=\frac{e^{-2.5}2.5^2}{2!} = 0.2565...$ P(X =3) $=\frac{e^{-2.5}2.5^3}{3!} = 0.2137...$ P(X =4) $=\frac{e^{-2.5}2.5^4}{4!} = 0.1336...$ P(X =5) $=\frac{e^{-2.5}2.5^5}{5!} = 0.0668...$

$= \sum_{i=0}^{5}(P(X = i)) = 0.957979$, 0.04202104 short of one due to the other κ values not accounted for e.g 6,7,8 (5+)

c) Explain why the 6 probabilities in part b) don't add up to 1.

The occurrence of one event does not affect the probability of a second event occurring, Events occur independently. The event of 5+ is not accounted for, Poisson distribution does not stop at 5, need to account for the other possibilities over 5.

```
Poissonprob = dpois(0:5,2.5) #probabilities from a Poisson distribution with mean=2.5
Poissonprob
```

```
## [1] 0.08208500 0.20521250 0.25651562 0.21376302 0.13360189 0.06680094
```

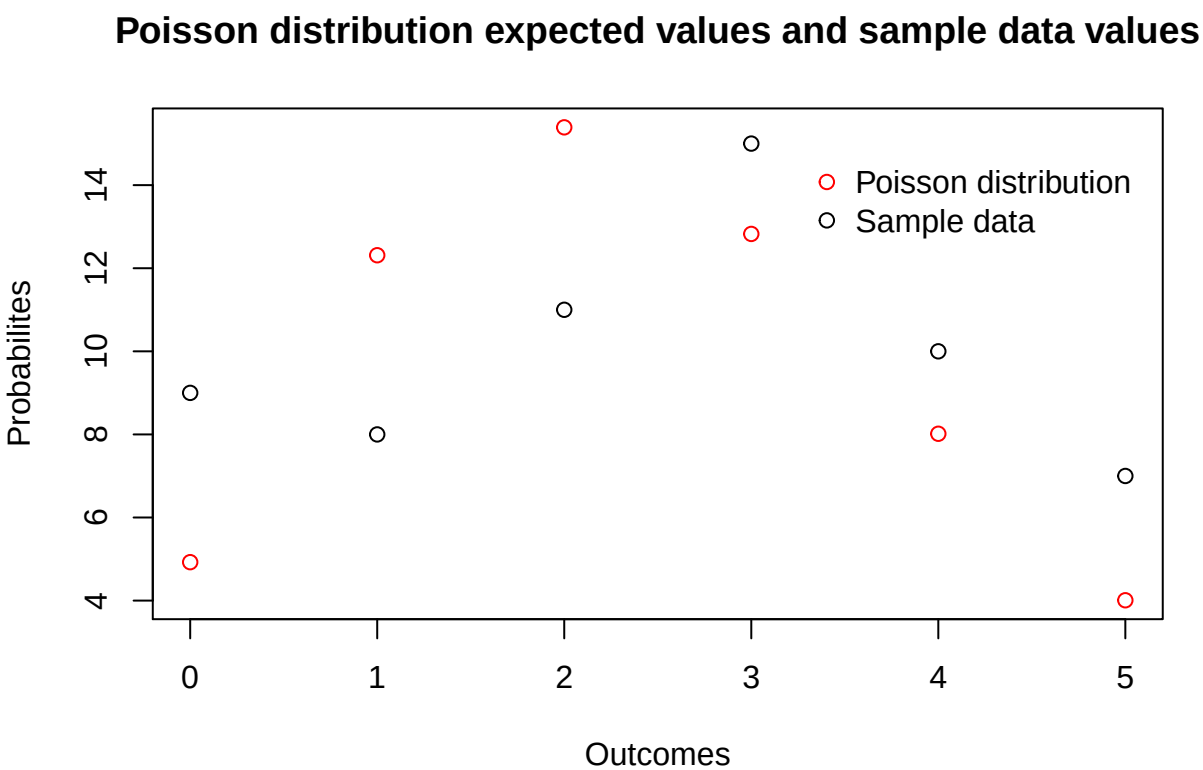
```
Sumofprob=sum(Poissonprob)
Sumofprob
```

```
## [1] 0.957979
```

```
1-Sumofprob
```

```
## [1] 0.04202104
```

```
#Expected values under poisson distribution
n0=(dpois(0,2.5)*60)#needs pooling
n1=(dpois(1,2.5)*60)
n2=(dpois(2,2.5)*60)
n3=(dpois(3,2.5)*60)
n4=(dpois(4,2.5)*60)
n5=(dpois(5,2.5)*60)#needs pooling, doesn't after adjusting poisson to include 5+ values
x1=c(0:5)
y1=c(n0,n1,n2,n3,n4,n5)
x2=c(0:5)
y2=c(9,8,11,15,10,7) #observed
plot(x1,y1, xlab = "Outcomes", ylab="Probabilites",
     main="Poisson distribution expected values and sample data values",
     type = "p", col="red")
points(x2,y2)
legend(3.25, 15, legend=c("Poisson distribution", "Sample data"),
       col=c("red","black"), pch=c(1,1), cex=1, bty = "n")
```




```
Exp=c(n0+n1,n2,n3,n4+n5) #first pool attempt
lastpart=60-sum(Exp)
NewExp=c(n0+n1,n2,n3,n4,n5+lastpart)
sum(NewExp)
```

```
## [1] 60
```

```
Obs=c(9+8,11,15,10,7) #new pooled observed values
X_sq <- sum((Obs- NewExp)^2/NewExp)
df <- (length(Obs) -1) - 1  #-1 as we have estimated the mean, -1 for parameter estimate
qchisq(0.95,df)
```

```
## [1] 7.814728
```

```
1- pchisq(X_sq, df) #?
```

```
## [1] 0.5419681
```

```
#Test for normality
shapiro.test(y2) #observed data
```

```
##
##  Shapiro-Wilk normality test
##
## data:  y2
## W = 0.92239, p-value = 0.5227
```

d)Test the hypothesis that the number of infected individuals in a household follows a Poisson distribution, taking into account the Poisson distribution probabilities dont add up to 1.

$$E_i = n\ddot{O}P(Pois(2.5) = i)$$

Using this, I can calculate and tabulate my expected values if following a Poisson Distribution.

Chi−squared test for goodness of fit H_0 : The number of infected individuals in a household follows a Poisson distribution. H_1 : The number of infected individuals in a household does not follows a Poisson distribution.

I realised infected 0 & 1 and 4 & 5 need to be pooled to satisfy the assumption of the test E_i<5. However the poisson looks at each stated values 0:5, the more than 5 value is not included, we can however calulate this easily using 1− sum of our probabilities of 0:5 or 60− summed frequency of expected values and add to the last bin making it equal to 1 and so we are able to use the X_2 test that has probability of 1 under distribution curve. Having done that I can now do a chisquared test on the new pooled data Obs and Exp with 3 degrees of freedom as I am estimating a parameter : $\hat{\boldsymbol{\theta}}$

Now

Under H_0:

$$X = \Sigma_{i,j} \frac{(O_{i,j} - E_{i,j})^2}{E_{i,j}} \approx X_d^2$$

d=m-1-1 (as estimating one parameter, the mean($\hat{\boldsymbol{\theta}}$))

Therefore we reject H_0 at the 5% level if X > $X_3^2(0.95)$ = 7.814728 For these data X = 2.149478 < $X_3^2(0.95)$ and so there is NO evidence to reject H0 at the 5% level. There is no evidence that the number of infected individuals in a household does not follows a Poisson distribution. This is as expected as the poisson model is an atypical model for disease outbreak and monitoring.[2] The p-value for this test was p = 0.5419681 > 0.05.

Assumptions:

1. Normally distributed sample data (Satisfied as p=0.5227 which is greater than 0.05, accepting H_0 at 5% level.)
2. Ei >5 (Satisfied after pooling)
3. Each observation is independent.

Question 3

Analysis of whether there are differences between the measured sample data to assess whether the purity of sodium chloride is evenly distributed through the barrel.

H_0: There are no differences between the samples in their purity H_1: At least one sample is different in ints purity Under H_0: F ~ $F_{n-1,n-t}$ so we reject H_0 at the 5% level if F>$F_{n-1,n-t}(0.95)$ with the degrees of freedom being n-1=4 and n-t=15

```
library(readr)
dframe1a<-read_csv("Purity.csv")
```

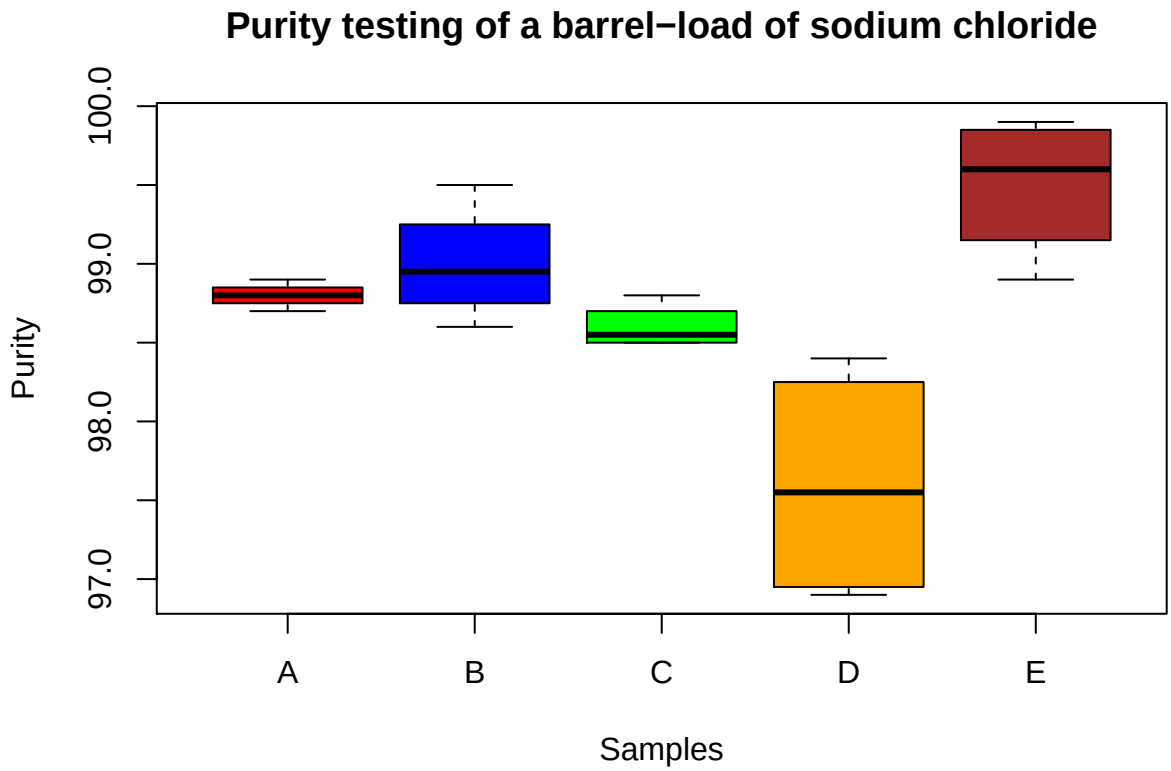
```
## Parsed with column specification:
## cols(
##   Sample = col_character(),
##   `Purity (%)` = col_double()
## )
```

```
dframe1a
```

```
## # A tibble: 20 x 2
##    Sample `Purity (%)`
##    <chr>         <dbl>
##  1 A              98.9
##  2 A              98.8
##  3 A              98.8
##  4 A              98.7
##  5 B              99.5
##  6 B              98.9
```

```
##  7 B              98.6
##  8 B              99
##  9 C              98.6
## 10 C              98.8
## 11 C              98.5
## 12 C              98.5
## 13 D              98.4
## 14 D              98.1
## 15 D              97
## 16 D              96.9
## 17 E              99.8
## 18 E              99.9
## 19 E              99.4
## 20 E              98.9
```

```
attach(dframe1a)
Samples<-tapply(`Purity (%)`, Sample, list)
boxplot(Samples, xlab="Samples", ylab="Purity", main="Purity testing of a barrel-load of sodium chlorid
```




```
mean(Samples$A)
```

```
## [1] 98.8
```

```
mean(Samples$B)
```

```
## [1] 99
```

```
mean(Samples$C)
```

```
## [1] 98.6
```

```r
mean(Samples$D)
```

```
## [1] 97.6
```

```r
mean(Samples$E)
```

```
## [1] 99.5
```

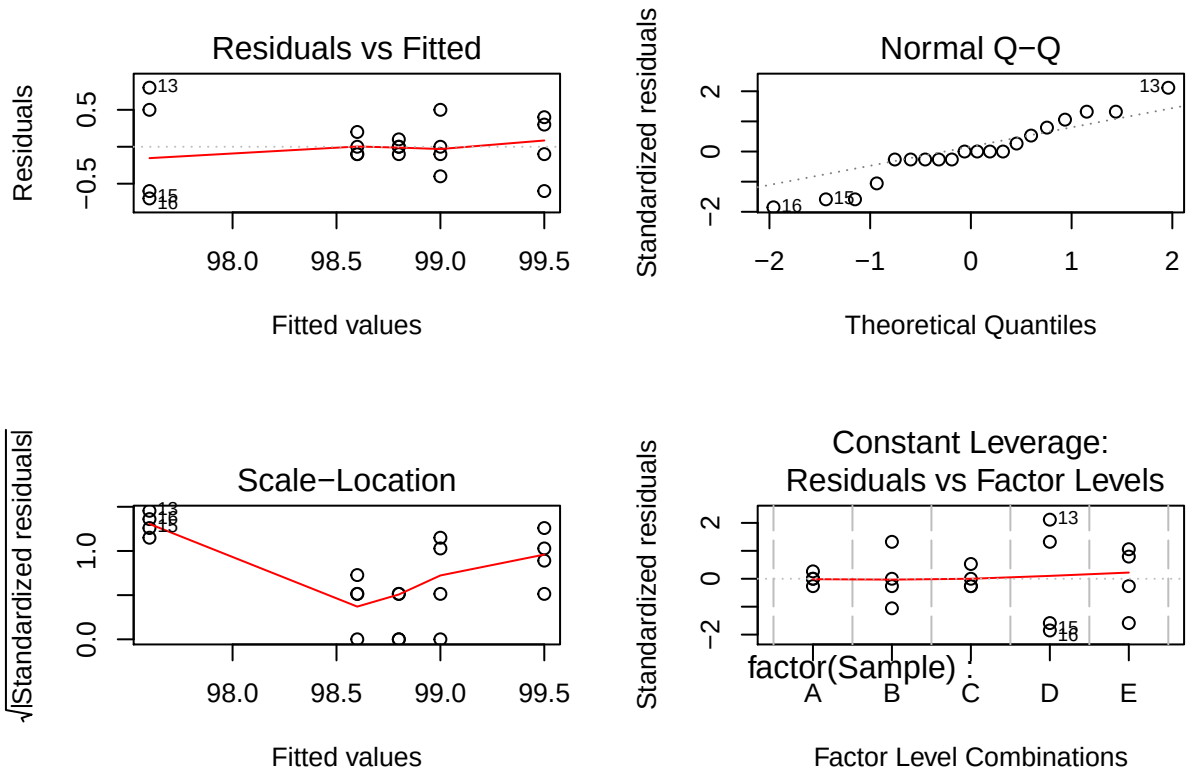
```r
anova1<-aov(`Purity (%)` ~ factor(Sample))
print(anova1)
```

```
## Call:
##    aov(formula = `Purity (%)` ~ factor(Sample))
##
## Terms:
##                 factor(Sample) Residuals
## Sum of Squares            7.84      2.86
## Deg. of Freedom              4        15
##
## Residual standard error: 0.4366539
## Estimated effects may be unbalanced
```

```r
anova=summary(anova1)
print(anova)
```

```
##                Df Sum Sq Mean Sq F value   Pr(>F)
## factor(Sample)  4   7.84  1.9600   10.28 0.000327 ***
## Residuals      15   2.86  0.1907
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
op<-par(mfrow=c(2,2))
plot(anova1)
```




```r
qf(0.95,4,15)
```

```
## [1] 3.055568
```

```r
1-pf(10.28, 4, 15)
```

```
## [1] 0.0003272696
```

```r
#Assumptions testing
bartlett.test(`Purity (%)` ~ factor(Sample)) #Variances not equal, quantifiable method
```

```
##
##  Bartlett test of homogeneity of variances
##
## data:  Purity (%) by factor(Sample)
## Bartlett's K-squared = 12.505, df = 4, p-value = 0.01397
```

```r
shapiro.test(Samples$A)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  Samples$A
## W = 0.94466, p-value = 0.683
```

```r
shapiro.test(Samples$B)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  Samples$B
## W = 0.96058, p-value = 0.7825
```

```r
shapiro.test(Samples$C)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  Samples$C
## W = 0.82743, p-value = 0.1612
```

```r
shapiro.test(Samples$D)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  Samples$D
## W = 0.84682, p-value = 0.216
```

```r
shapiro.test(Samples$E)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  Samples$E
## W = 0.91647, p-value = 0.5174
```

From the plot we can see equal variances with the widths of the plots being the same, it further shows sample D being lower than the other samples and sample E being slightly higher than the others. Since this p-value is 0.000327<0.05 there is evidence to reject the H_0 at the 5% significance level that all of the samples are the same.

Our F value test statistic (F=10.28) is past the threshold of 3.055568 ($F_{4,15}(0.95)$) at the 5% confidence interval. Therefore the purity of sodium chloride is shown not to be evenly distributed through the barrel.

Based on the data I believe this is true with differences in the means of the samples A:E being evidently different: E>B>A>C>D [A] 98.8 [B] 99 [C] 98.6 [D] 97.6 [E] 99.5 but the low number of repeats would prove to make this unreliable.

Some of the anova assumptions are sattisfied: Relationship of residual to fitted values is shown to be linear with the plot(anova1) therefore showing equal variances with the points being spread out equally but not quantifiable enough to state this. Bartlett test quantifiably shows that this is does not have equal variances as pvalue of $0.01397 < 0.05$ therefore reject H_0 :

H_0: The variances are equal in all the samples H_1: The variances are not equal in all the samples p-value = $0.01397 < 0.05$ shows that the variances are not equal. (Not homogenous)

The qq plot showing normality, however slight skew, a shapiro test for normality is more of a quantitative test for normality and shows a p value of $0.02387 > 0.05$ so confirms normality.This was repeated for each sample and showed normality each time, If wasnt normal would have to transform data via root, square, log functions or use a non-parametric Kruskal Wallis test. Observations are independent as one sample does not affect the other and from different areas however could be not independent as the barrel samples are from the same barrel and salt may come from the same mine.

Question 4

An appropriate model for the number of chest infections experienced by patients during the study.

As the data of interest is count data, a simple linear model would not be suitable however a generalised linear model for counts is appropriate. Generation of firstly all the factors in a GLM is useful for determination of the suitable predictors.

```r
library(readr)
dframe22a<-read_csv("ChestInfections.csv")
```

```
## Warning: Missing column names filled in: 'X17' [17]
```

```
## Parsed with column specification:
## cols(
##   ChestInfections = col_double(),
##   EnteredStudy = col_character(),
##   LeftStudy = col_character(),
##   Gender = col_character(),
##   Age = col_double(),
##   BMI = col_double(),
##   WeightInKg = col_double(),
##   HeightInCm = col_double(),
##   SystolicBP = col_double(),
##   DiastolicBP = col_double(),
##   EyeColour = col_character(),
##   Smoker = col_double(),
##   Asthmatic = col_double(),
##   SalbutamolInhaler = col_double(),
##   `Days in Study` = col_double(),
##   EyeColProportion = col_double(),
##   X17 = col_double()
## )
```

```r
# I made a new column of days in study as it is a factor that is important to be tested in future.
attach(dframe22a)
dframe22a$`Days in Study` #shows the new data column
```

```
##  [1]  945 1010  926  918 1054  777  917  920 1065 1052  781  764 1049  938
## [15]  992  763  713  923  886  703 1020 1054  822  833 1023 1043  888  764
## [29]  951  913 1047 1001  846  743 1043  784  902  942  830
```

```r
GLMall<-glm(formula = ChestInfections~Gender+Age+BMI+WeightInKg+HeightInCm+SystolicBP
            +DiastolicBP+EyeColour+Smoker+Asthmatic+SalbutamolInhaler+`Days in Study`,
            family = poisson(link="log"))
summary(GLMall)
```

```
##
## Call:
## glm(formula = ChestInfections ~ Gender + Age + BMI + WeightInKg +
##     HeightInCm + SystolicBP + DiastolicBP + EyeColour + Smoker +
##     Asthmatic + SalbutamolInhaler + `Days in Study`, family = poisson(link = "log"))
##
## Deviance Residuals:
##      Min        1Q    Median        3Q       Max
## -0.91821  -0.48546  -0.15108   0.05428   2.03049
##
## Coefficients:
##                      Estimate Std. Error z value Pr(>|z|)
## (Intercept)        -6.292e+01  4.362e+01  -1.442   0.1492
## Gendermale         -1.593e-01  6.375e-01  -0.250   0.8027
## Age                 1.603e-02  4.399e-02   0.364   0.7155
## BMI                 7.537e-01  7.545e-01   0.999   0.3178
## WeightInKg         -2.743e-01  3.054e-01  -0.898   0.3691
```

```
## HeightInCm         3.116e-01  2.488e-01   1.253   0.2104
## SystolicBP         5.486e-02  4.207e-02   1.304   0.1922
## DiastolicBP        2.461e-03  7.637e-02   0.032   0.9743
## EyeColourbrown     1.965e+00  9.336e-01   2.105   0.0353 *
## EyeColourgreen    -1.319e+01  3.468e+03  -0.004   0.9970
## EyeColourgrey      2.121e+00  1.330e+00   1.594   0.1108
## Smoker             2.301e+00  5.599e-01   4.110 3.96e-05 ***
## Asthmatic          3.266e+00  1.549e+00   2.109   0.0350 *
## SalbutamolInhaler -8.623e-01  1.227e+00  -0.703   0.4822
## `Days in Study`   -2.147e-03  2.568e-03  -0.836   0.4032
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 84.050  on 38  degrees of freedom
## Residual deviance: 12.506  on 24  degrees of freedom
## AIC: 84.614
##
## Number of Fisher Scoring iterations: 15
```

```r
GLMesa<-glm(formula = ChestInfections~EyeColour+Smoker+Asthmatic,family = poisson(link="log"))
summary(GLMesa)
```

```
##
## Call:
## glm(formula = ChestInfections ~ EyeColour + Smoker + Asthmatic,
##     family = poisson(link = "log"))
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -1.6298  -0.6642  -0.6527   0.4069   1.8646
##
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -1.54646    0.43881  -3.524 0.000425 ***
## EyeColourbrown    0.03498    0.37639   0.093 0.925961
## EyeColourgreen  -14.75612 2103.36257  -0.007 0.994402
## EyeColourgrey     0.98388    0.52096   1.889 0.058948 .
## Smoker            1.79522    0.37217   4.824 1.41e-06 ***
## Asthmatic         1.29846    0.38617   3.362 0.000773 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 84.050  on 38  degrees of freedom
## Residual deviance: 34.488  on 33  degrees of freedom
## AIC: 88.596
##
## Number of Fisher Scoring iterations: 14
```

```r
GLMeye<-glm(formula = ChestInfections~EyeColour, family = poisson(link="log"))
summary(GLMeye) #Shows the eye colours are not significant
```

```
##
## Call:
## glm(formula = ChestInfections ~ EyeColour, family = poisson(link = "log"))
##
## Deviance Residuals:
##    Min      1Q  Median      3Q     Max
## -2.000  -1.414  -1.283   0.423   3.391
##
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -7.111e-17  2.425e-01   0.000    1.000
## EyeColourbrown -1.942e-01  3.609e-01  -0.538    0.591
## EyeColourgreen -1.530e+01  1.276e+03  -0.012    0.990
## EyeColourgrey   6.931e-01  4.287e-01   1.617    0.106
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 84.050  on 38  degrees of freedom
## Residual deviance: 78.396  on 35  degrees of freedom
## AIC: 128.5
##
## Number of Fisher Scoring iterations: 13
```

```r
cor.test(ChestInfections,SalbutamolInhaler)
```

```
##
##  Pearson's product-moment correlation
##
## data:  ChestInfections and SalbutamolInhaler
## t = 2.4083, df = 37, p-value = 0.02113
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.0595113 0.6124940
## sample estimates:
##       cor
## 0.3681163
```

```r
GLMallsal<-glm(formula = ChestInfections~Gender+Age+BMI+WeightInKg+HeightInCm+SystolicBP
               +DiastolicBP+EyeColour+Smoker+Asthmatic+`Days in Study`,family = poisson(link="log"))
summary(GLMallsal)
```

```
##
## Call:
## glm(formula = ChestInfections ~ Gender + Age + BMI + WeightInKg +
##     HeightInCm + SystolicBP + DiastolicBP + EyeColour + Smoker +
##     Asthmatic + `Days in Study`, family = poisson(link = "log"))
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -1.0185  -0.4571  -0.1192   0.0487   2.0568
##
## Coefficients:
##                   Estimate Std. Error z value Pr(>|z|)
## (Intercept)     -5.791e+01  4.143e+01  -1.398  0.16223
## Gendermale      -7.713e-02  6.209e-01  -0.124  0.90114
```

```
## Age              1.191e-02  4.333e-02   0.275  0.78336
## BMI              6.320e-01  6.952e-01   0.909  0.36335
## WeightInKg      -2.221e-01  2.801e-01  -0.793  0.42790
## HeightInCm       2.728e-01  2.319e-01   1.176  0.23949
## SystolicBP       6.239e-02  4.053e-02   1.540  0.12368
## DiastolicBP     -5.590e-03  7.694e-02  -0.073  0.94209
## EyeColourbrown   1.748e+00  8.987e-01   1.945  0.05176 .
## EyeColourgreen  -1.297e+01  3.468e+03  -0.004  0.99702
## EyeColourgrey    1.993e+00  1.274e+00   1.565  0.11762
## Smoker           2.276e+00  5.560e-01   4.094 4.24e-05 ***
## Asthmatic        2.360e+00  7.999e-01   2.951  0.00317 **
## `Days in Study` -1.251e-03  2.285e-03  -0.547  0.58420
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 84.050  on 38  degrees of freedom
## Residual deviance: 12.993  on 25  degrees of freedom
## AIC: 83.101
##
## Number of Fisher Scoring iterations: 15
```

```r
anova(GLMallsal, GLMall, test="LRT")
```

```
## Analysis of Deviance Table
##
## Model 1: ChestInfections ~ Gender + Age + BMI + WeightInKg + HeightInCm +
##     SystolicBP + DiastolicBP + EyeColour + Smoker + Asthmatic +
##     `Days in Study`
## Model 2: ChestInfections ~ Gender + Age + BMI + WeightInKg + HeightInCm +
##     SystolicBP + DiastolicBP + EyeColour + Smoker + Asthmatic +
##     SalbutamolInhaler + `Days in Study`
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1        25     12.993
## 2        24     12.506  1  0.48673   0.4854
```

```r
GLMsa<-glm(formula = ChestInfections~Smoker+Asthmatic,family = poisson(link="log"))
summary(GLMsa)
```

```
##
## Call:
## glm(formula = ChestInfections ~ Smoker + Asthmatic, family = poisson(link = "log"))
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -1.8897  -0.7209  -0.7209   0.3982   1.9669
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.3477     0.3448  -3.908 9.30e-05 ***
## Smoker        1.9273     0.3676   5.243 1.58e-07 ***
## Asthmatic     1.0107     0.3211   3.147  0.00165 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 84.050  on 38  degrees of freedom
## Residual deviance: 38.977  on 36  degrees of freedom
## AIC: 87.085
##
## Number of Fisher Scoring iterations: 5
```

```r
GLMs<-glm(formula = ChestInfections~Smoker,family = poisson(link="log"))
summary(GLMs)
```

```
##
## Call:
## glm(formula = ChestInfections ~ Smoker, family = poisson(link = "log"))
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -2.2962  -0.8451  -0.8451   0.8296   1.8988
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.0296     0.3162  -3.256  0.00113 **
## Smoker        1.9990     0.3667   5.451 5.01e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 84.050  on 38  degrees of freedom
## Residual deviance: 48.417  on 37  degrees of freedom
## AIC: 94.525
##
## Number of Fisher Scoring iterations: 6
```

```r
GLMa<-glm(formula = ChestInfections~Asthmatic,family = poisson(link="log"))
summary(GLMa)
```

```
##
## Call:
## glm(formula = ChestInfections ~ Asthmatic, family = poisson(link = "log"))
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -2.0548  -1.1547  -1.1547   0.1513   3.3886
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.4055     0.2236  -1.813 0.069786 .
## Asthmatic     1.1527     0.3204   3.598 0.000321 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
```

```
##     Null deviance: 84.050  on 38  degrees of freedom
## Residual deviance: 71.874  on 37  degrees of freedom
## AIC: 117.98
##
## Number of Fisher Scoring iterations: 6
```

```r
#Nested tests
anova(GLMsa, GLMs, test="LRT")
```

```
## Analysis of Deviance Table
##
## Model 1: ChestInfections ~ Smoker + Asthmatic
## Model 2: ChestInfections ~ Smoker
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1        36     38.977
## 2        37     48.417 -1  -9.4403 0.002123 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(GLMsa, GLMa, test="LRT")
```

```
## Analysis of Deviance Table
##
## Model 1: ChestInfections ~ Smoker + Asthmatic
## Model 2: ChestInfections ~ Asthmatic
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1        36     38.977
## 2        37     71.874 -1  -32.898 9.714e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
#all factors vs all factors-salbutomol
#Non nested tests
AIC(GLMs,GLMa) #non-nested test , AIC shows asthma and smoker are similar power predictors
```

```
##      df       AIC
## GLMs  2  94.52507
## GLMa  2 117.98237
```

```r
BIC(GLMs,GLMa) #non-nested test , AIC shows asthma and smoker are similar power predictors
```

```
##      df      BIC
## GLMs  2  97.8522
## GLMa  2 121.3095
```

```r
GLMsaOff<-glm(formula = ChestInfections~offset(EyeColProportion)+Asthmatic+Smoker,family = poisson(link=
summary(GLMsaOff)
```

```
##
## Call:
## glm(formula = ChestInfections ~ offset(EyeColProportion) + Asthmatic +
##     Smoker, family = poisson(link = "log"))
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -1.9565  -0.7399  -0.7399   0.3998   2.3831
##
## Coefficients:
```

```
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.7316     0.3441  -5.032 4.86e-07 ***
## Asthmatic     0.9452     0.3213   2.942  0.00326 **
## Smoker        1.9448     0.3678   5.288 1.24e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 86.386  on 38  degrees of freedom
## Residual deviance: 41.733  on 36  degrees of freedom
## AIC: 89.841
##
## Number of Fisher Scoring iterations: 5
```

```r
odds_Asth <- exp(coefficients(GLMa)["Asthmatic"]) #determination of odds with just asthma glm
odds_Asth
```

```
## Asthmatic
##  3.166667
```

```r
odds_Asth2 <- exp(coefficients(GLMsaOff)["Asthmatic"]) #determination of odds with final choosen glm
odds_Asth2
```

```
## Asthmatic
##  2.573207
```

```r
GLMallOff<-glm(formula = ChestInfections~Gender+offset(EyeColProportion)+Age+BMI+WeightInKg
               +HeightInCm+SystolicBP+DiastolicBP+Smoker+Asthmatic+SalbutamolInhaler+
                 `Days in Study`,family = poisson(link="log"))
summary(GLMallOff)
```

```
##
## Call:
## glm(formula = ChestInfections ~ Gender + offset(EyeColProportion) +
##     Age + BMI + WeightInKg + HeightInCm + SystolicBP + DiastolicBP +
##     Smoker + Asthmatic + SalbutamolInhaler + `Days in Study`,
##     family = poisson(link = "log"))
##
## Deviance Residuals:
##      Min        1Q    Median        3Q       Max
## -1.51705  -0.49008  -0.18127   0.03173   1.69531
##
## Coefficients:
##                     Estimate Std. Error z value Pr(>|z|)
## (Intercept)        4.8900260 21.9003165   0.223    0.823
## Gendermale         0.1011207  0.4908190   0.206    0.837
## Age                0.0584837  0.0371159   1.576    0.115
## BMI               -0.3306492  0.3655818  -0.904    0.366
## WeightInKg         0.1717349  0.1464135   1.173    0.241
## HeightInCm        -0.0850619  0.1209775  -0.703    0.482
## SystolicBP         0.0217184  0.0297046   0.731    0.465
## DiastolicBP       -0.0490377  0.0797072  -0.615    0.538
## Smoker             2.4742252  0.5058840   4.891    1e-06 ***
## Asthmatic          1.4713417  1.1598212   1.269    0.205
## SalbutamolInhaler  0.1836666  1.0311257   0.178    0.859
```

```
## `Days in Study`    0.0008992  0.0022757   0.395    0.693
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 86.386  on 38  degrees of freedom
## Residual deviance: 18.736  on 27  degrees of freedom
## AIC: 84.845
##
## Number of Fisher Scoring iterations: 6
```

```
pvalues=summary(GLMallOff)$coefficients[2:12, 4]
pvaluessort<-sort(pvalues,decreasing = FALSE) #has to be in ascending order
HolmBonferoni<-c(0.05/12,0.05/11,0.05/10,0.05/9,0.05/8,0.05/7,0.05/6,0.05/5,
                 0.05/4,0.05/3,0.05/2,0.05/1)
HolmBonferoni
```

```
##  [1] 0.004166667 0.004545455 0.005000000 0.005555556 0.006250000
##  [6] 0.007142857 0.008333333 0.010000000 0.012500000 0.016666667
## [11] 0.025000000 0.050000000
```

```
as.numeric(pvaluessort)
```

```
##  [1] 1.003786e-06 1.150941e-01 2.045861e-01 2.408179e-01 3.657587e-01
##  [6] 4.646903e-01 4.819801e-01 5.384075e-01 6.927296e-01 8.367718e-01
## [11] 8.586268e-01
```

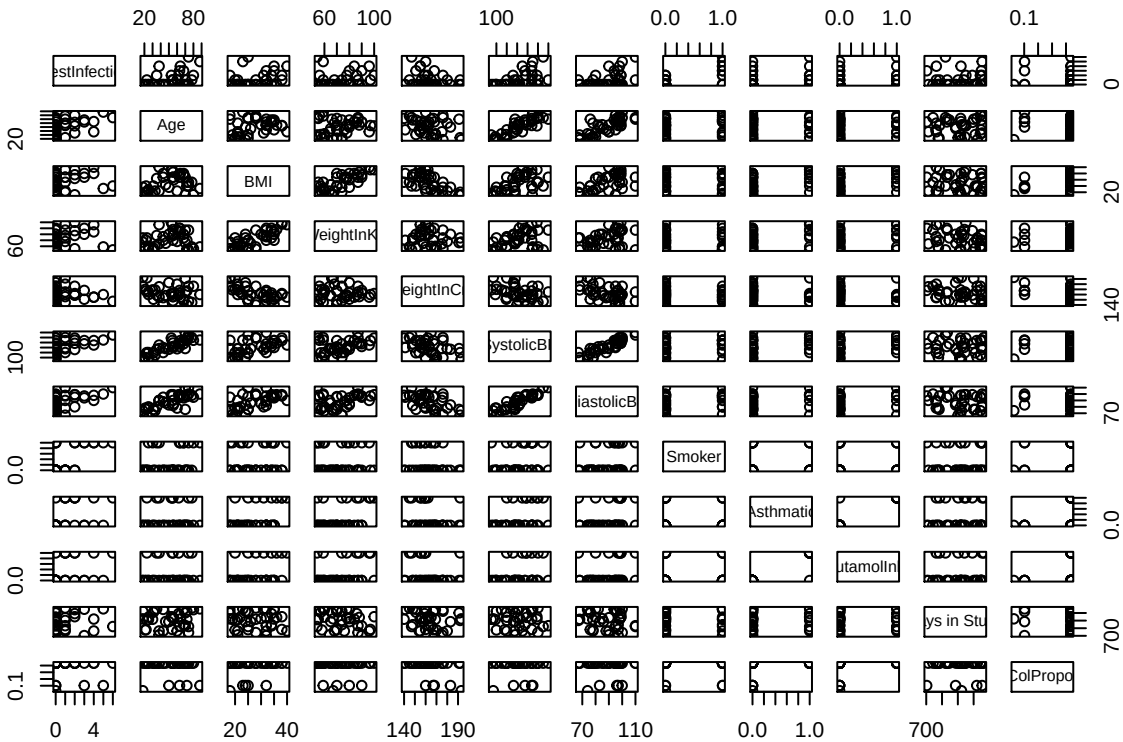
```
pvaluesadj<- p.adjust(pvaluessort, method=c("fdr"), n=length(pvaluessort))
true.discoveries <- length(which(pvaluesadj[1:10]<=0.05))
true.discoveries #only shows one (smoker)
```

```
## [1] 1
```

```
# This shows that the p value shows only the smoking factor needs to be counted in the model.
dframe23a<-read_csv("ChestInfectionsNum.csv") #numerical data frame
```

```
## Parsed with column specification:
## cols(
##   ChestInfections = col_double(),
##   Age = col_double(),
##   BMI = col_double(),
##   WeightInKg = col_double(),
##   HeightInCm = col_double(),
##   SystolicBP = col_double(),
##   DiastolicBP = col_double(),
##   Smoker = col_double(),
##   Asthmatic = col_double(),
##   SalbutamolInhaler = col_double(),
##   `Days in Study` = col_double(),
##   EyeColProportion = col_double()
## )
```

```r
pairs(dframe23a)  #correlation visual for salbutimol as a cause
```



Explanation of chosen model, including model assumptions, and justification of choices made.

As I wanted to reduce the chance of identifying a lifestyle factor as important when it wasn't in my model, I want to control the familywise error rate. The p-values in a regression model are dependent, and so the assumptions for a Sidak correction are not satisfied. I would now perform a Holm-Bonferroni correction as this is more powerful for this dataset as it doesnt assume independence. It reports only the smoker needs to be included but however its assumption of no independence may make it slighlty less appropriate as some predictors may be independent. Attempts to adjust the p values using the False discovery rate correction still proves to yield one (smoker) as the only significant factor. This is further shown in the GLM all with offset as smoker as the only important factor. Smoking has shown to be such a strong predictor, it therefore paramount to get the experimental set up of this factor quantitativley and appropriately as discussed bellow.

a) The most significant risk factor for getting chest infections:

Smoking has the lowest and most significant Wald p value (3.96e-05) according to the GLMall model for all predictors. Further after fitting a GLM for just the significant predictors, eye colour was removed as not signifcant (Wald p value) and further models of Smoking, Asthma and Smoking+Asthma were generated. I further then used the Likelihood ratio test to find out which was the best model between the nested models listed above. I found out Smoking to still remain the best with an LRT p value of (9.714e-09) compared to the LRT p value for asthma (0.002123). Similar BIC and AIC values for the smoking and asthma test show they are both valid predictors: 94, 118 for AIC and 98 121 for BIC for smoking and asthma respectively.

b) Does a Salbutamol Inhaler cause chest infections?

No but Asthma does, salbutimol inhaler use has been shown in the GLMall to be an insignificant factor to chest infections (insignificant p value for Wald 0.4822), people who have asthma use salbutimol inhalers however this correlation does not mean causation. An LRT test confirms this as shown by p value of 0.4854 for the test of salbutimol. Further a correlation test of Chest Infections to Salbutimol shows a weak correlation of 0.3681163, even though this weak correlation does not mean causation.

c) How many times as many chest infections would an asthmatic get than a non-asthmatic?

Using this now determined predictor of asthma and smoking being important predictors, I made a glm of just astmatic as a predictor and found it leads to 3.166667 more likely Chest.Infections using the expenential of the intercept as it gives the odds of success when all the predictors are set to 0.. However after looking at my offset in part D and generating a glm with offset and including smoking and asthma as significant predictors, I found out it is 2.573207 times more likely. I went for the latter as my choice of model.

d) Appropriate method for eye colour modelling in the dataset

We want to use a Poisson GLM, with EyeColour-proportion as an offset which we dont need to estimate a regression coefficient. This is as an offset as it is important to take into account differences in the size of the samples, i.e. the population of brown, gree, blue and grey eye colour is different and will thus expect different cases of chest infections per group because of the difference in proportions. I generated a proportions eye colour column by counting the frequency and divding by the total number of individuals and addressed this in the dataframe. I further modelled it in the Asthma and Smoking GLM as an offset.

e) How could the design of future experiments be improved to ensure that the inferences were appropriate for both smokers and non-smokers? Frequency of smoking, how often they smoke , quantifiable differences in Smoking =1 to non smoking =0 so can see the differences. Design of experiments. Replacement of the categorical qualitative data to quantitatative data with previous smoking history noted as well.

Conclusions: Final model of Asthmatic, Smoker and offest eyecolour proportion is the strongest GLM in my judgement as it includes al the important factors, not just the strongest smoker predictor and further accounts for population diffferences in eye colour proportion as an offset rather than a predictor.

Assumptions fo the GLM :

1. The response has a distribution from the exp family.
2. The response is determined by the predictors in a linear model fashion.
3. The mean of the response is connected to the predictors through a link function (log).

##Bibliography 1.G. Weaving, G. Batstone and R. Jones, Annals of Clinical Biochemistry, 2016, 53, 106-111. 2.L. Held, Biostatistics, 2005, 7, 422-437.